

Supporting information

Effect of confinement of TiO₂ nanotubes over the Ru nanoparticles on Fischer-Tropsch synthesis

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Table S1 FTS performance of Ru-in-TNT reacted in various temperatures.

Temp. (°C)	^a Conver.	^b Select. (%)		
	(%)	CH ₄	C ₂₋₄	C ₅₊
200	25	20.6	15.0	64.4
220	32	22.7	14.4	62.9
240	44	22.4	15.6	62.0
260	61	24.5	12.0	63.5

^a CO conversion rate at the initial reaction period^b Selectivity determined at similar CO conversion of 20% for each run**Table S2** The specifications of fresh catalysts.

Catal.	^a Mean particle size (nm)	^b Metal dispersion	^b Surface Ru sites (mmol g ⁻¹)	^c TOFs (10 ⁻³ s ⁻¹)
Ru-out/TNT	3.2	0.41	0.204	20
Ru-in/TNT	2.8	0.47	0.224	36

^a average particle size calculated from the TEM image by recording randomly selected 200 particles .^b the metal dispersion and surface Ru sites were determined using equation (1) and (2), respectively[1].

$$D=1.32/d \quad (1)$$

$$[\text{Ru}]_{\text{surf}}=[\text{Ru}]^0 \times D \quad (2)$$

Where the D , d , $[\text{Ru}]_{\text{surf}}$ and $[\text{Ru}]^0$ stand for the metal dispersion, mean particle diameter, surface Ru sites and total amount of Ru, respectively.

^c Evaluated from the rate of CO conversion at the initial 2h reaction period per Ru atom on the surface

Table S3 The specifications of spend catalysts.

Catal.	Mean particle size (nm)	Metal dispersion	Surface Ru sites (mmol g ⁻¹)	^a TOFs (10 ⁻³ s ⁻¹)
Ru-out/TNT	5.5	0.24	0.028	17
Ru-in/TNT	4.0	0.33	0.134	31

^a Evaluated from the rate of CO conversion at 24 h reaction per Ru atom on the surface

Table S4 The amount of Ru in the spend catalysts and liquid phase of product.

Catalyst	^a Ru concentration (mmol g ⁻¹)		
	C_{bulk}	C_{liquid}	C_{Total}
Ru-out/TNT	0.12	0.35	0.47
Ru-in/TNT	0.41	0.08	0.49

^a The C_{bulk} and C_{liquid} were referred as the Ru concentration in the solid bulk of used catalyst, in the liquid phase of FTS product, respectively, which were determined by the ICP technique. The C_{Total} was defined as the sum of C_{bulk} and C_{liquid} .